BOUBAKAR DIALLO

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EDUCATION

Cornell University — **B.S. Electrical & Computer Engineering** — **May 2028**

Honors: Community Leadership Award - Cornell University (Aug 2025)

Coursework: Machine Learning, Foundations of Robotics, Embedded Systems, Control Systems, Computer Architecture, Signals & Systems, Machine Learning, Linear Algebra

SKILLS

Programming & Robotics: Python, C/C++, ARM Assembly, MATLAB, Bash, ROS/ROS2, Pixhawk/ArduPilot, PID Control, GPS Waypoint Navigation, Sensor Fusion (Radar/IMU/LiDAR), NVIDIA Jetson Orin Nano

Hardware & Tools: PCB Design & Bring-Up, Motor Drivers, ESCs, Power Distribution, Sensor Calibration, Oscilloscopes, Logic Analyzers, MATLAB, Git/GitHub

Embedded Systems: ARM Cortex-M, RTOS, Real-Time Scheduling, Interrupts, I^2C, SPI, UART, GPIO

WORK EXPERIENCE

Embedded Systems Intern — *ADAS SAFE, Jacksonville, FL* — **Jun – Aug 2025**

- Validated and integrated automotive radar, IMU, and LiDAR hardware for driver-assistance systems, diagnosing sensor calibration dependencies and reducing radar alignment errors by 15% across production deployments.
- Root-caused microsecond-level sensor synchronization drift during perception-hardware bring-up, characterizing inter-sensor timing dependencies that informed sensor-fusion hardware/software interfaces.
- Automated diagnostic and reliability-test workflows using statistical process control across multimodal sensor streams, detecting calibration drift and sensor misalignment early.

Robotics & Hardware Systems Lead — *Engineers Without Borders, Ithaca, NY* — **Oct 2024 – Jan 2026**

- Designed and fabricated custom PCBs, motor-driver circuits, and power-regulation subsystems for an agricultural rover adapted from NASA's open-source Mars rover architecture, integrating a ROS-controlled robotic arm.
- Deployed CNN inference on NVIDIA Jetson Orin Nano under embedded Linux, applying transfer learning and spatial augmentation to improve crop-disease detection accuracy by 40% with GPIO-triggered alerts.
- Integrated autonomous UAV systems with GPS-guided waypoint navigation and custom image-capture pipelines, developing Python test sequences to validate motor control and navigation across three field sites.

Software Engineer Intern — *Simulacrum, NYC, NY* — **Jan – Jun 2026**

- Engineered a distributed real-time sensor-data platform using FastAPI and Google Cloud Pub/Sub, processing high-throughput time-series streams with sub-100 ms latency for forecasting and model-training workflows.
- Built observability pipelines with structured logging and automated alerts for latency spikes, data drift, and processing failures, reducing debugging cycles by approximately 30%.

Incoming Technology Intern — *PNC Bank* — **Summer 2027**

PROJECTS

Autonomous Mobile Robot — Electrical & Controls — **Jan – Sep 2026**

- Developed closed-loop PID motor control with sensor feedback for autonomous navigation, integrating actuator interfaces, signal-conditioning circuits, and embedded control logic.
- Designed DC/DC power regulation, battery management, and custom electrical interfaces, validating hardware performance under load using oscilloscopes and multimeters.

Hexacopter UAV — *Pixhawk, ArduPilot* — **Jul 2026**

- Built and integrated a six-rotor UAV with ESC motor-drive harnesses, power-distribution hardware, GPS/compass, and Pixhawk flight controller, enabling live voltage/current telemetry.
- Calibrated accelerometer, compass, ESC throttle ranges, and RC endpoints; configured RC-loss and low-battery failsafes and validated the system through bench and tethered testing.

Preemptive RTOS & Real-Time Scheduler — *C, ARM Assembly, Cortex-M0+* — **Dec 2025**

- Developed a preemptive RTOS kernel implementing context switching, process-state queues, spin/blocking locks, and Earliest Deadline First scheduling.
- Implemented 1 ms interrupt-driven timing, deadline-prioritized ready queues, and preemption of non-real-time processes, validated concurrency and synchronization through custom stress tests.

Bidirectional Room Occupancy Monitor — *C, ARM Cortex-M0+, I^2C, SPI* — **May 2026**

- Built a bidirectional people counter using dual VL53L0X time-of-flight sensors on a shared I^2C bus, sequencing sensor initialization to resolve address conflicts.
- Implemented a two-zone entry/exit state machine, MAX7219 LED-matrix output over SPI, and UART event logging to a Python application.